ANDROID PROJECT PROPOSAL

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Submitted By:

Name: Azwa Jamal & Iqra Maqsood Mughal

Roll No: 110846 & 110847

Course: Android App Development

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Project: WhatsApp Clone with Bluetooth Offline Feature

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Signature: _____

WhatsApp Clone with Bluetooth Offline Feature

Introduction

This proposal outlines the development of a WhatsApp-like chat application that works both online and offline. When inte:rnet is available, the app works like normal WhatsApp using Firebase. When internet is not available, the app automatically switches to Bluetooth mode and allows users to chat with nearby devices without any internet connection. This hybrid approach ensures users can always communicate, whether they have internet access or not.

Project Objectives:

- Develop a WhatsApp clone that works like the original WhatsApp when internet is available.
- Enable users to send and receive messages without internet using Bluetooth.
- Add automatic mode switching between online and offline modes.
- Provide secure and reliable chat experience in both modes.
- Support real-world scenarios where internet is unreliable or unavailable.

Scope of Work

User Management

- User registration and login using phone number with OTP verification.
- Profile setup with name, profile picture, and about status.
- Auto-login feature for returning users.

Online Mode Features (When Internet Available)

- One-to-one text messaging like WhatsApp.
- Group chat creation and management.
- Send images, videos, and voice notes.
- Read receipts showing single tick, double tick, and blue ticks.
- Typing indicator showing when someone is typing.
- Push notifications for new messages.
- Call history and status stories like WhatsApp.

Offline Mode Features (When Internet Not Available)

- App automatically detects when internet is unavailable.
- Bluetooth mode activates automatically.
- Nearby users within Bluetooth range are discovered and listed.
- Text messages can be sent and received via Bluetooth.
- Messages automatically relay through nearby devices using mesh network.
- When internet returns, all offline messages sync with Firebase automatically.

Bluetooth Feature Details

- Users within 10 to 30 meters can chat directly.
- Messages can hop through intermediate devices to reach farther users.
- Only text messages work in Bluetooth mode due to speed limitations.
- Users can see who is nearby and available for chat.
- Connection status shows whether in online or offline mode.

Limitations

- Bluetooth mode range is limited by device hardware, typically 10 to 30 meters.
- Photos, videos, and voice notes cannot be sent in Bluetooth mode, only text messages.
- Bluetooth mode is slower than internet mode.
- Both users must have the app installed and Bluetooth on for offline chat.
- Firebase free plan supports only 100 simultaneous users, which is fine for 75 users.

Technology Stack

Frontend (Android App)

- Java programming language is used for development.
- XML is used for UI design.
- Android Studio is used as the development environment.

Online Mode Backend

- Firebase Authentication is used for phone number login.
- Firebase Firestore is used for storing messages online.
- Firebase Storage is used for photos, videos, and voice notes.
- Firebase Cloud Messaging is used for push notifications.

Offline Mode Connectivity

- Bluetooth API is used for device discovery and messaging when internet is off.
- WiFi Direct API is used as an alternative for faster transfer.

Local Storage

- Room Database is used to store offline messages when internet is not available.
- Shared Preferences are used for saving user session and settings.

Security

- AES encryption is used for message transmission in both modes.
- Messages are encrypted before sending via Bluetooth.
- Firebase provides built-in security for online mode.

Security Measures

- Messages are encrypted before transmission in both online and offline modes.
- User authentication is done through Firebase phone number verification.
- Chat history is stored securely in Firebase when online and locally when offline.
- No message is sent in plain text, even over Bluetooth.
- User privacy is maintained as no cloud storage is used for offline messages.

System Features

- Dual-mode operation switches automatically between online and offline.
- User-friendly interface similar to WhatsApp.
- Real-time message updates in online mode.
- Bluetooth-based messaging in offline mode with nearby devices.
- Automatic mode detection without user intervention.
- Message queue system stores messages when offline and sends when internet returns.
- Group chat works in online mode only.
- Notifications work in both modes when possible.
- Profile pictures and status visible in online mode.
- Read receipts work in online mode.

Step by Step working:

- When user opens the app, it checks for internet connection.
- If internet is available, normal WhatsApp mode starts using Firebase.

- User can send text, photos, videos, and voice notes.
- Messages are delivered instantly through Firebase cloud.
- If internet is not available, Bluetooth mode activates automatically.
- App scans for nearby devices with the same app.
- User sees list of nearby people within Bluetooth range.
- User can only send text messages in this mode.
- Messages are sent directly from one phone to another via Bluetooth.
- If direct connection is not possible, messages relay through other phones.
- When internet comes back, all offline messages sync to Firebase.
- User does not need to do anything; mode switching is automatic.

Development Methodology

- Requirement analysis is done first to understand all online and offline needs.
- UI/UX design is created to match WhatsApp interface.
- Development phase includes coding online features with Firebase.
- Bluetooth module is developed separately for offline mode.
- Mode switching logic is implemented to detect internet status.
- Testing is done with internet on and off to verify both modes.
- Integration testing ensures messages are not lost during mode switching.
- Deployment is done on target devices for 75 users.
- Maintenance and updates are provided as needed.

Expected Outcomes

- A fully functional WhatsApp clone will work like original WhatsApp when internet is available.
- Bluetooth offline feature will allow text messaging when internet is not available.
- Automatic mode switching will work seamlessly without user intervention.
- All 75 users will be able to communicate in both online and offline modes.
- Messages will not be lost during mode switching; they will sync when internet returns.
- Complete Java source code will be provided with documentation.

Target Users

- Students in schools and universities can use this app for campus communication.
- Event organizers in remote areas can coordinate without internet.
- Users in emergency scenarios can communicate when networks are down.

- People in areas with poor internet connectivity can still chat via Bluetooth.
- Any group of up to 75 people who need reliable communication in all conditions.

Comparison with Normal WhatsApp

Feature	Normal WhatsApp	This App
Online messaging	Yes	Yes
Photos and videos	Yes	Yes
Voice notes	Yes	Yes
Group chat	Yes	Yes
Read receipts	Yes	Yes
Offline messaging	No	Yes with Bluetooth
Internet required	Always	Only for media and sync
Range limitation	Worldwide	Bluetooth range for offline

Conclusion

This project is a WhatsApp clone with an added Bluetooth feature for offline communication. When internet is available, the app works exactly like WhatsApp using Firebase for all features. When internet is not available, the app automatically switches to Bluetooth mode and allows users to send text messages to nearby devices. This hybrid approach ensures that 75 users can always communicate with each other regardless of internet availability. The app is ideal for schools, events, emergency situations, and areas with poor internet connectivity.